CO2-AT1 Analytical problem solving

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1 Covariance Analysis in health insurance

premium increase (%) x	claims per 100 policyholders (y)
3.2	15
4.0	18
5.5	25
4.8	20

Sol :- * calculate the mean :-

$$\text{Mean of } \bar{x} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4}$$

$$= \frac{17.5}{4} = 4.375 \quad \therefore \boxed{\bar{x} = 4.375}$$

$$\text{Mean of } \bar{y} = \frac{15 + 18 + 25 + 20}{4}$$

$$= \frac{78}{4} = 19.5 \quad \therefore \boxed{\bar{y} = 19.5}$$

* calculate deviations and their products :-

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3.2	15	−1.175	−4.5	5.2875
4.0	18	−0.375	−1.5	0.5625
5.5	25	1.125	5.5	6.1875

-* Smooth ...

4 8	2 0	0.425	0.5	0.2125
				=12.25

$\Sigma (x-\bar{x})(y-\bar{y}) = 5.2275 + 0.5625 + 6.1875 + 0.2125$
$= 12.25$

* Calculate Covariance :-

formula, $Cov(x,y) = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{n}$

here, $\Sigma (x-\bar{x})(y-\bar{y}) = 12.25$

$n = 4$

$Cov(x,y) = \frac{12.25}{4} = 3.0625$

$\therefore$ mean of $x = 4.375$

mean of $y = 19.5$

Covariance $= 3.0625$

2 Data smoothing using equal frequency Binning:

Sol:- Given data

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 128
130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

* Divide the Data into equal frequency :-

Bin	Value
Bin1	110, 112, 115, 115, 118, 120, 120, 122, 124
Bin2	124, 126, 126, 126, 130, 135, 135, 138
Bin3	138, 138, 138, 140, 145, 150, 152, 160, 168

* Smooth the data using Bin value

Bin1 :-

$$\text{Mean} = \frac{110+112+115+118+120+120+115+122+124}{9}$$

$$= \frac{1056}{9} = 117.33$$

Bin2 :-

$$\text{Mean} = \frac{124+126+126+126+126+130+135+135+136}{9}$$

$$= \frac{1166}{9} = 129.56$$

Bin3 :-

$$\text{Mean} = \frac{138+138+138+140+145+150+152+160+180}{9}$$

$$= \frac{1341}{9} = 149$$

Smoothed Bin3: 149, 149, 149, 149, 149, 149, 149, 149, 149 //

3. Equal-width Binning for Risk categorization

Given data (sorted BMI value)

18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Step1 :- find the Bin width

minimum BMI = 18.5

maximum BMI = 39.5

Range = 39.5 - 18.5 = 21

Number of Bins = 3

Bin width = $\frac{21}{3} = 7$

therefore, the bins are:

Bin1 : 18.5 – 25.5

Bin2 : 25.5 – 32.5

Bin3 : 32.5 – 39.5

step2 :- partition the data

Bin	values
Bin1 (18.5–25.5)	18.5, 19.2, 21.8, 24.9, 25.5
Bin2 (25.5–32.5)	26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4
Bin3 (35.5–39.5)	33.1, 33.6, 34.7, 38.2, 39.5

step3 :- smooth by bin boundaries and replace each values

Bin	original values	Boundary – smoothed values
Bin1	18.5, 19.2, 21.8, 24.9, 25.5	18.5, 18.5, 18.5, 25.5, 25.5
Bin2	26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4	25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5
Bin3	33.1, 33.6, 34.7, 38.2, 39.5	32.5, 32.5, 32.5, 39.5, 39.5

the BMI values are partioned into three equal width bins and each values is replaced with the nearest bin boundary. the smoothing technique reduces variation while the preserving the overall distribution of the data

4. Data Reduction using Bin medians (Medical supply costs)

Given data (Cost in ₹)

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155,
165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 350

step1:- sort the data:-

120, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 160, 210, 230, 350.

there are 27 values so, divide them into 3-equal frequency bins, each containing 9 values

step2:- divide into equal frequencies:-

Bin	Values	Median
Bin 1	120, 140, 145, 145, 148, 150, 150, 150, 155	148
Bin 2	155, 155, 155, 160, 160, 165, 170, 170, 175	160
Bin 3	175, 175, 175, 180, 190, 200, 210, 230, 350	190

step3:- Replace each value with bin median

Bin1:- smoothed:-

148, 148, 148, 148, 148, 148, 148, 148

Bin2:- smoothed:-

160, 160, 160, 160, 160, 160, 160, 160, 160

Bin 3:- ~~smoo~~ Original:-

175, 175, 175, 180, 190, 200, 210, 230, 350

smoothed:-

190, 190, 190, 190, 190, 190, 190, 190, 190

5 standard deviation of patient Age Data

Given data :-

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

(a) Compute the mean absolute deviation (MAD)

step1 :- Calculate mean

$$\bar{x} = \frac{18+22+25+42+28+43+33+35+56+28}{10}$$

$$= \frac{330}{10} = 33$$

Mean Age $\bar{x} = 33$

step2 :- Calculate absolute deviations

Age (x)	$x-\bar{x}$
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

$x-\bar{x} = 15+11+8+9+5+10+0+2+23+5 = 88$

steps :- Calculate MAD

$$MAD = \frac{88}{10} = 8.8$$

$\therefore$ mean absolute deviation $= 8.8$

(b) compute the z-score for the first four observation

step 1:-

standard deviation $\sigma = \sqrt{\frac{\Sigma (x-\bar{x})^2}{n}}$

$= \sqrt{\frac{1174}{10}} = \sqrt{117.4} \cong 10.84$

step 2:- calculate z-score

$$z = \frac{x-\bar{x}}{\sigma}$$

(i) $x-\bar{x} = 0-5$

$z = \frac{-5}{10.84} = -1.38$

(ii) $x-\bar{x} = -11$

$z = \frac{-11}{10.84} = -1.02$

(iii) $x-\bar{x} = -8$

$z = \frac{-8}{10.84} = -0.74$

(iv) $x-\bar{x} = 9$

$z = \frac{9}{10.84} = 0.83$